Let X denote the number of families purchasing Hindu Paper

p = Probability of their family purchasing the Hindu

$$p = \frac{40}{200} = \frac{1}{5}$$

$$q = \frac{4}{5}$$

$$n = 10$$

(i) Only one family purchase the Hindu

$$P(X=1) = nC_1\, p^1\, q^{n-1}$$

$$= 10C_1 \times \frac{1}{5} \times \left(\frac{4}{5}\right)^9$$

$$= 2 \times \left(\frac{4}{5}\right)^9$$

(ii) No family purchasing the Hindu

$$P(X=0) = 10C_0 (\frac{1}{5})^0 (\frac{4}{5})^{10}$$

$$= (\frac{4}{10})^{10}$$

(iii) Not more than one family purchasing The Hindu means that $X \leq 1$

$$P(X \leq 1) = P[x=0] + P[x=1]$$

$$= 10C_0 (\frac{1}{5})^0 (\frac{4}{5})^{10} + 10C_1 (\frac{1}{5})^1 (\frac{4}{9})^9$$

$$= (\frac{4}{5})^{10} + 10 \times (\frac{1}{5})^1 (\frac{4}{5})^9$$

$$= (\frac{4}{5})^9 \left[(\frac{4}{5}) + 2\right]$$

$$= (\frac{4}{5})^9 (\frac{14}{5})$$

Example 10.9

In a tourist spot, 80% of tourists are repeated visitors. Find the distribution of the numbers of repeated visitors among 4 selected peoples visiting the place. Also find its mode or the maximum visits by a visitor.

Solution:

Let the random variable X denote the number of repeated visitors.

$X \sim B\ (n, p)$